

Chandan Kumar Singh

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Summary

Backend Software Engineer with 1+ year of experience designing and delivering high-performance Java/Spring Boot microservices. Proven track record of reducing API latency by 40-50%, cutting inter-service overhead by 60%, and eliminating security incidents through AES-256 encryption and JWT-based authentication. Experienced in building cloud-native systems on AWS, Docker, and Kubernetes while owning end-to-end delivery of scalable APIs and event-driven architectures.

Experience

Tata Consultancy Services

Software Engineer — Java, Spring Boot, Microservices

Jan 2025 – Present

- **API Gateway & System Design:** Architected a centralized API Gateway with secret-key authentication and JWT token-reuse strategy, cutting inter-service overhead by 60% and boosting distributed system reliability.
- **Scalable API Design:** Designed and owned RESTful APIs serving business-critical banking workflows; reduced average response latency by 40% and improved platform scalability by 50% through service-layer optimizations.
- **Security Engineering:** Engineered secure integrations with external banking systems using AES-256 encryption and defensive patterns against MITM, Replay, and Request Tampering attacks, achieving zero security incidents.
- **Automation & Ownership:** Owned end-to-end delivery of internal automation solutions, eliminating 64% of manual operational effort and accelerating team release velocity.
- **High-Throughput Engineering:** Built a Bulk Upload pipeline for concurrent ingestion of thousands of records per request, increasing system throughput by 35% while maintaining zero data-consistency issues.

Projects

Clothify Shopping Application | Java, Spring Boot, Hibernate, PostgreSQL, React.js, gRPC, Kafka, Docker, Kubernetes, AWS

- **Distributed Microservices Design:** Built a scalable e-commerce backend with API Gateway routing, Eureka service discovery, and LLD-driven domain separation across product, user, and order services.
- **Event-Driven & Cloud-Native Deployment:** Engineered low-latency gRPC + Kafka-based async communication with JWT security; containerized and deployed on Kubernetes (AWS EC2) via Jenkins CI/CD, sustaining 100+ concurrent users with 30% latency reduction.

Technical Skills

Languages: Java, Python, C++, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Security, Hibernate, REST API, gRPC, Kafka, FastAPI, React.js

Databases: PostgreSQL, MySQL, Redis, FAISS, ChromaDB

Cloud: AWS (EC2, S3), Docker, Kubernetes, Jenkins

Tools: Git, GitHub, SVN, Jenkins, NumPy, Pandas, Scikit-learn, HuggingFace, Transformers, LangChain

Education

Motihari College of Engineering, Motihari

B.Tech in Computer Science and Engineering (CSE)

Oct 2020 – Aug 2024